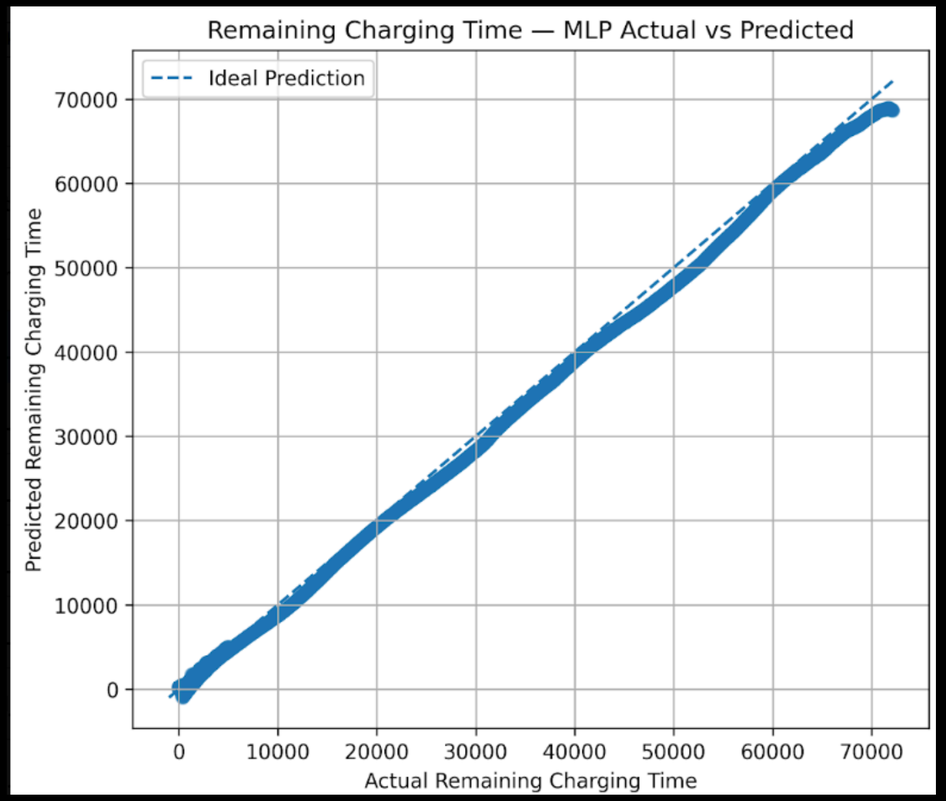
Battery SOC & Charging Time Prediction

NAME:- SREEJA SANYAL
WEEK :-4
DAY:-28
DATE:- 30.10.2026

Project Overview

- Worked with lithium-ion battery charging data.
- Used machine learning to estimate battery State of Charge (SOC).
- Predicted the Remaining Charging Time.



Dataset and Feature

- Used battery charging data containing voltage, current, temperature, capacity, and other measurements.

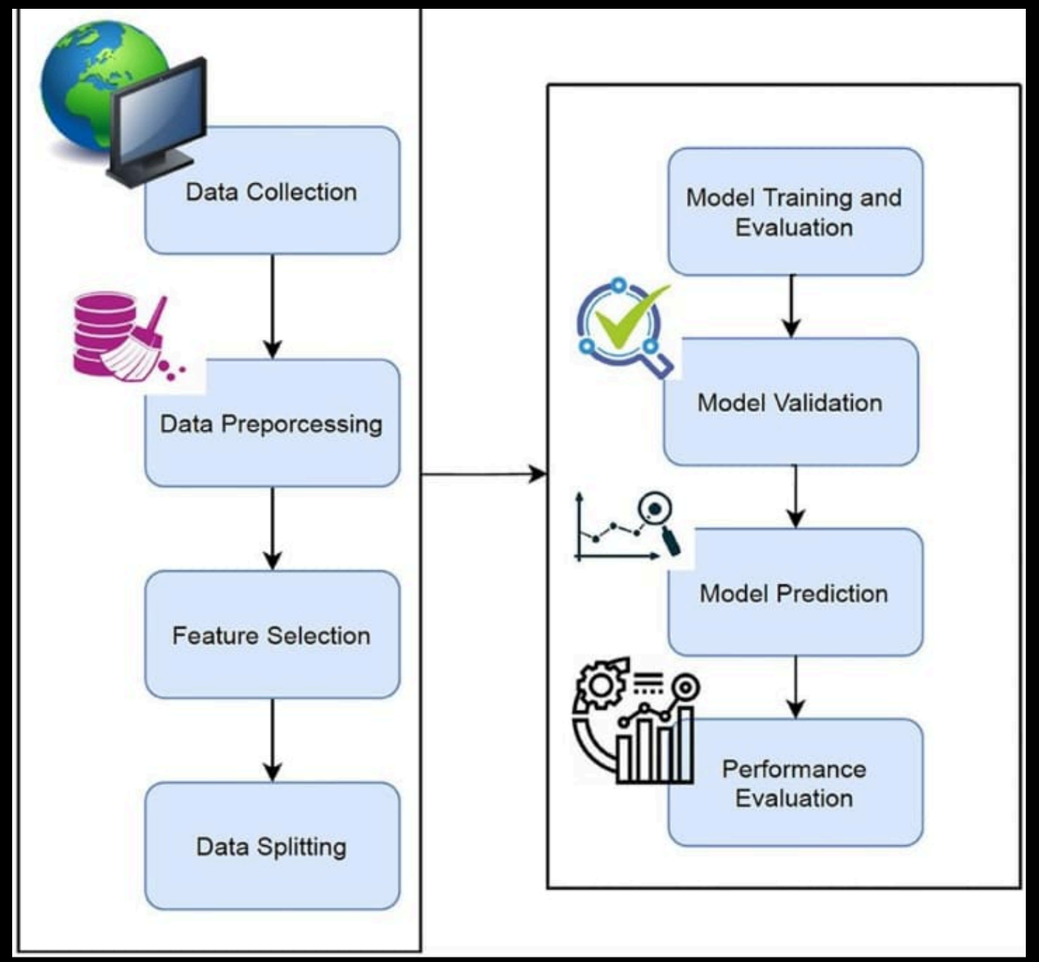
- Final model features: Voltage, Current, Temperature, Elapsed Time, Cycle, WhAccu, and Power.

```
SOC raw features:
['Voltage', 'Current', 'Temperature', 'Elapsed Time', 'Cycle', 'WhAccu']

Remaining-time raw features:
['Voltage', 'Current', 'Temperature', 'Elapsed Time', 'Cycle', 'Capacity', 'WhAccu']
```

Methodology

- Data collection
- Data preprocessing
- Feature selection
- Model training
- Model evaluation
- Performance analysis



Models Used

- XGBoost Regressor: SOC estimation.
- Random Forest Regressor: Remaining Charging Time prediction.

```
FINAL MODEL RESULT SUMMARY
SOC Prediction
Model: XGBoost Regressor
Test R²: 0.990521
Remaining Charging Time Prediction
Model: Random Forest Regressor
Test R²: 0.998756
Remaining Charging Time Validation
Model: Random Forest Regressor
Validation R²: 0.996502
```

Validation performnce comparison

- SOC Estimation: XGBoost Regressor was selected based on its validation R^2 performance.
- Remaining Charging Time: Random Forest Regressor achieved the best validation R^2 among the tested models and was selected as the final model.

Task	Final Model	R2
SOC Prediction	XGBoost Regressor	0.990521
Remaining Charging	Random Forest Regr	0.998756
Remaining Charging	Random Forest Regr	0.996502

Software Artefacts

- Project notebooks
- Saved trained models
- Preprocessing and prediction workflow
- Evaluation outputs and plots
- Project source files

Task	Final Model	R2
SOC Prediction	XGBoost Regressor	0.990521
Remaining Charging	Random Forest Regr	0.998756
Remaining Charging	Random Forest Regr	0.996502

Fig:- overall comparison table



Fig:-saved models

Limitations

- Prediction errors may occur under certain charging conditions.
- Performance may vary with temperature and operating conditions.
- Dataset coverage can affect model performance.
- The models may not capture every complex pattern in battery charging behaviour.
- Prediction accuracy may be affected by variations in battery ageing and degradation.

Conclusion

The project developed machine learning models to estimate battery SOC and predict Remaining Charging Time, with results on the available dataset.

Thank You